26 accident-derived defects, real tool output per language
C
MISRA C
Rust
Ada
SPARK
Spelf
M-01 Unguarded f64 → i16 narrowing (Ariane 501)
M-02 Unit swap lbf·s ↔ N·s (Mars Climate Orbiter)
M-03 Frame swap Body ↔ ECI
M-04 Signed 32-bit centisecond counter overflow (787 GCU)
M-05 Epoch mix MET ↔ TAI (Deep Impact)
M-06 Buffer over-read trusting a length field
M-07 CCSDS length field: forgotten '+1'
M-08 Hand-written wrong ceiling priority (Pathfinder)
M-09 Unprotected state shared by two tasks
M-10 Saturated measurement used directly (Schiaparelli)
M-11 Last good value persists without age limit
M-12 Transient latched without debounce (MPL)
M-13 'Queue full' result ignored (Spirit, sol 18)
M-14 Backup channel shares primary's code (Ariane 501)
M-15 Halt on fault during flight phase (Ariane 501)
M-16 Out-of-range configuration constant (Titan IVB)
M-17 Error quaternion not canonicalised (long way round)
M-18 Clamped command used blind (integrator windup)
M-19 Reuse outside the verified envelope (Ariane 501)
M-20 Safe-mode path depends on failed gyro
M-21 Recursion without declared bound (stack depth)
M-22 Uninitialised read
M-23 Division by a computed zero
M-24 Implicit rounding in fixed point
M-25 Three channels, unsigned link, one Byzantine channel
M-26 Second spike passes filter at hold expiry (QF72)
✓ static, unconditional
◆ static, needs a declaration
R caught only at run time
✗ not caught